Youth Drug Usage Data Analysis

DATA 5322 Homework 1

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Agenda

Introduction

- Research question
- Why youth drug use matters?
- Models that will be used

Theoretical Background

- Decision trees & Ensemble Methods
- Classification vs Regression

Methodology

- Data cleaning & feature engineering
- Model Setup(tree, RF, boosting)

Results

- Model Performance comparison
- Key predictors

Discussion

- Tree interpretation
- Data type Comparison(binary vs ordinal vs numeric)
- Ethical Consideration

Conclusion

- Key findings
- Limitations and future work

Research Question(s)

The Research questions are:

- Whether a youth has ever used any drug(binary classification- Used/Never Used).
- The frequency of marijuana use in the past year (multi-class classification).
- The number of days per year a youth has used marijuana(regression) ?

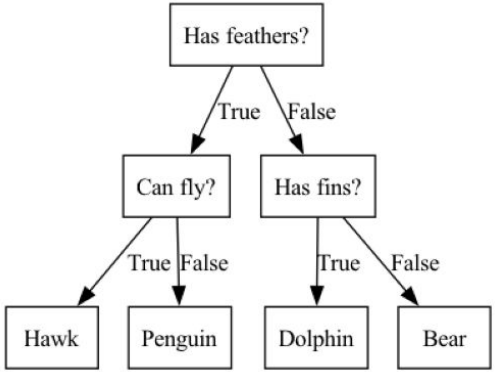
The tree and ensemble approaches that will be used:

- Decision Tree Approach
- Ensemble approach(Bagging, Boosting, Rain Forest)

Theoretical Background & Methodology

Decision Tree Ensemble Methods

- Recursively splits data into purer subgroups using the predictor and threshold that best separates outcome classes.
- Decisions are made until a stopping criterion is reached.
- Very easy to interpret - can trace prediction down to leaf.
- Tends to overfit training data - small changes often result in different trees.
- Can implement using tree() function or rpart() function in R.
- rpart() allows for more control over parameters.
- Split data into training and test, fit tree using training data, prune tree if too complex, use pruned tree for prediction on test set.
- Evaluated with precision, recall, accuracy score, error rate, f1-score, ROC curves and AUC across all classification thresholds.



Theoretical Background & Methodology

Ensemble Approaches - Bagging

- Bagging, also known as Bootstrap Aggregation.
- An ensemble method which involves training multiple models independently on random subsets of data, and aggregating their prediction through voting(for classification) or averaging(for aggregation).
- Considers all predictors at each split(mtry = p).
- Bagging can be done using randomForest() function in R.
- Predictions are made by splitting dataset into smaller groups based on rules.

Theoretical Background & Methodology

Ensemble Approaches - Random forest

- An ensemble method which uses many trees to make better predictions.
- Each tree looks at different random parts of data and their results are combined by voting of classification or averaging for regression.
- Considers only subset of predictors at every split(mtry=p/3 for regression; mtry = sqrt(p) for classification.
- It can be done using randomForest() function in R.

Theoretical Background & Methodology

Ensemble Approaches - Boosting

- An ensemble method which uses many trees to make better predictions.
- Rather than building trees in parallel, Boosting builds trees sequentially.
- Key hyperparameter - shrinkage rate controls how each tree contributes to the final model and precents overfitting.
- Smaller shrinkage rate require more complex trees but produces better model.
- It can be done using gbm() function in R.
- Find the the optimal shrinkage value that minimizes the error
- Set distribution = “guassian” for regression, distribution = bernouilli for classification.
- Set n.trees, interaction.depth, set shrinkage to the chosen value

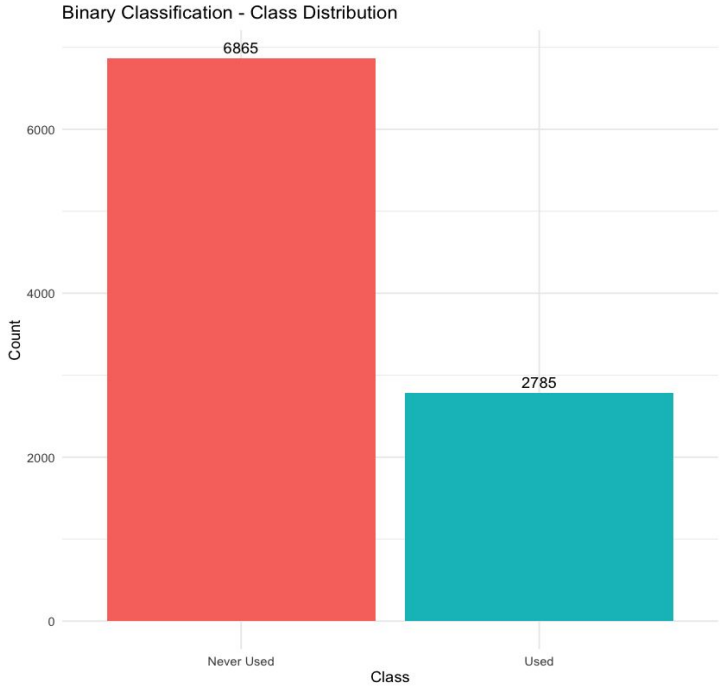
Methodology

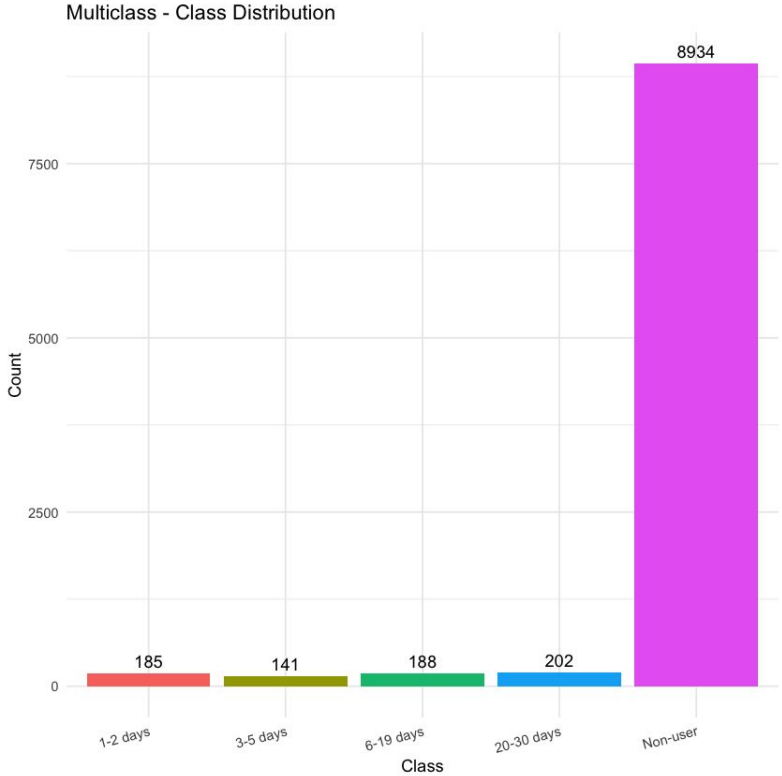
Data Preparation

- Data was cleaned by handling missing value and recoding special survey codes (e.g. 991, 993) as NA.
- Categorical and numerical variable were transformed to meaningful, interpretable formats
- Predictors of interest were selected :
  - Demographics
  - Parental Influence
  - Religiousity
- Variables were renamed for clarity and easier interpretation

Methodology

Handling Class Imbalance





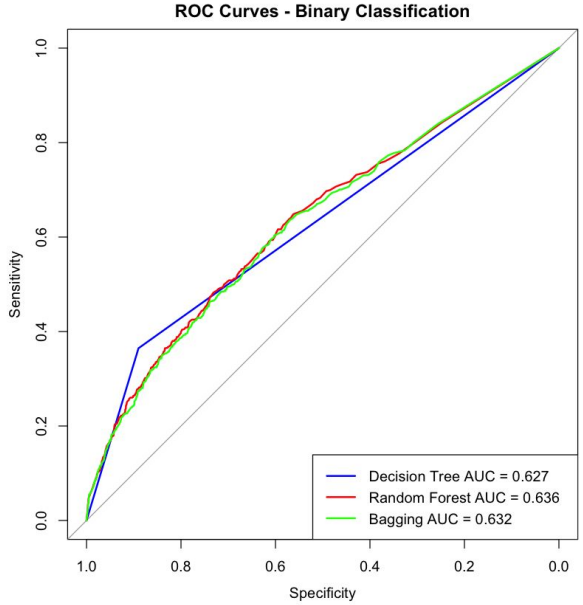
Methodology

Model Implementation - Tuning and Validation

- Splitting dataset into 70:30 split (training and testing)
- Decision trees - searched for best combination of cp and max tree depth using minimum error.
- For random forest and bagging - searched for best combination mtry and ntrees.
- For boosting, searched for optimal shrinkage value, number of trees and interaction depth.
- Picked parameters that minimized test error or MSE.
- Models were then refitted using these optimal parameters and predictions were made.

Results

Binary Classification Problem - Decision



A data.frame: 3 × 2

Model	f1_score
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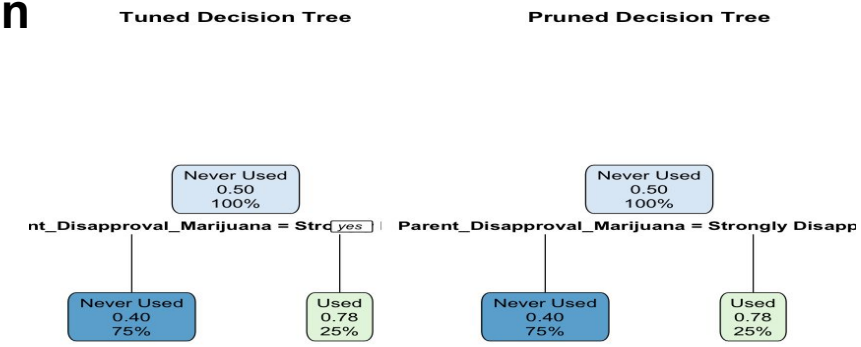
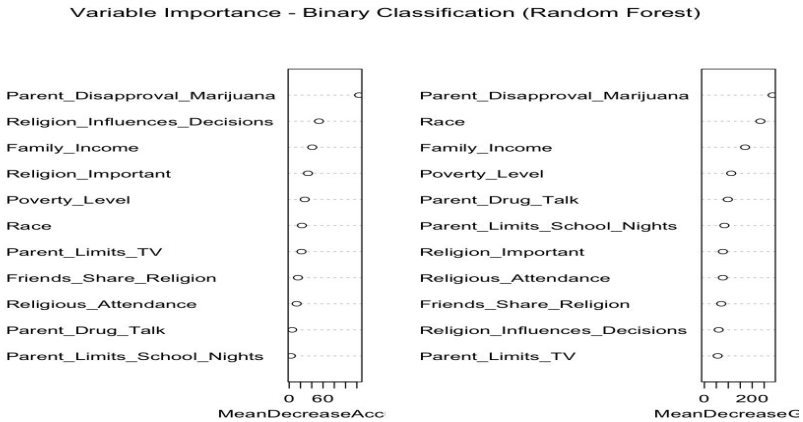
Decision Tree	0.4465455
Random Forest	0.3735119
Bagging	0.3694362





Results

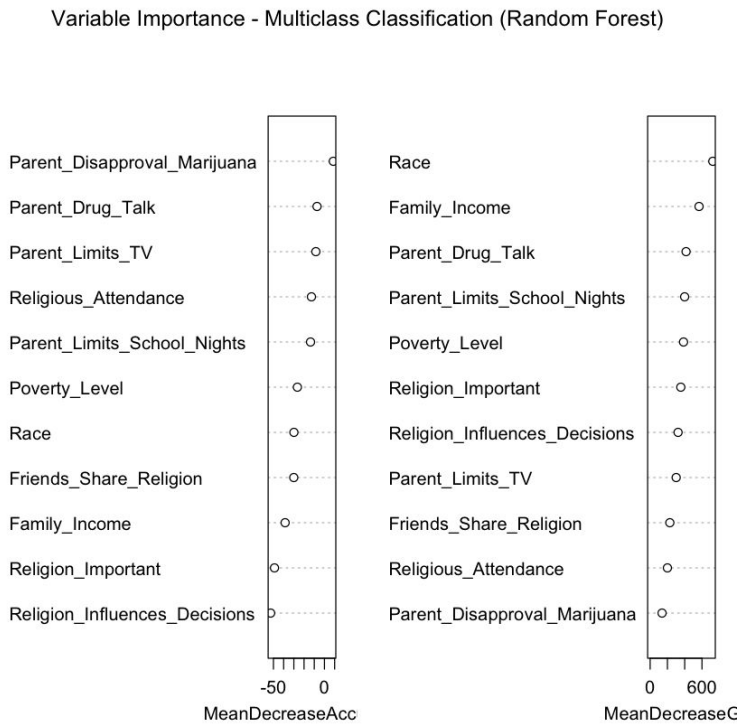
Multi-Class Classification Problem - Random Forest

The variables of most importance in predicting if a youth consumes drug include:

- Parent_Disapproval_Marijuana(PRMJEVR2)
- Poverty_Level(POVERTY3)
- Family_Income(INCOME)
- Religion_important (RLGDCSN - how important religion is to an individual's decision)

F1 scores:

- 1-2 days: ~0.024
- 3-5 days: ~0.056
- 6-19 days: ~0.0345
- 20-30 days: ~0.122
- Non user: ~ 0.802



Results

Regression Problem - Random Forest & Boosting

Comparing the MSE for both models:

- Boosting model and Random Forest obtained similar RMSE(111 vs 112) .

- This suggests the predictors available in this dataset — demographics, parental attitudes, and religiosity — capture some signal but are far from sufficient to precisely predict marijuana use frequency.
- Important predictors include:
  - Parent_Disapproval_Marij
  - Religion_important
  - Religion_Influences_Decisions
  - Family Income
  - Poverty Level

Discussion

What are the implications of this outcome? How can you, as the data science communicator, discuss these findings in an ethical way?

- Parental disapproval of marijuana — strongest predictor across all models
- Religious engagement — religion influences decisions, importance of beliefs
- Socioeconomic factors — family income and poverty level
- Demographics — weakest and least consistent predictor

Discussing findings:

- Strong parental disapproval is associated with lower likelihood of drug use
- Religious engagement acts as an additional protective factor
- These are associations not causal relationships — other unmeasured factors such as peer influence and mental health may also be at play

Discussion

What are the implications of this outcome?

- Prevention strategies should focus on strengthening family communication about drugs
- Religious and community networks may play a supporting role in reducing substance use
- Socioeconomic support may reduce risk factors associated with drug use

How can you, as the data science communicator, discuss these findings in an ethical way?

- Don’t assume causations as models just find patterns.
- Don’t use findings to generalize about any particular group.
- Be upfront about the limitations of datasets(i.e. Class imbalance)

Discussion

Some of these variables are the same information coded into binary, ordinal (categorical and ordered), and numerical variables. How do the predictions change using each data type? What is each telling you, and when is it appropriate to use each?

- **Binary (0/1)**
  - Focuses on whether a factor matters (yes/no)
  - Predictions change in one step (on vs off)
  - Best when only presence/absence is relevant
- **Numerical**
  - Captures full variation and potential thresholds
  - Predictions change continuously with values
  - Best when magnitude and differences are meaningful
- **Ordinal (Ordered Categories)**
  - Captures ranked levels (e.g., low → medium → high)
  - Predictions change stepwise across levels
  - Best when order matters but spacing is unclear
-

Conclusion

Most important predictors:

- Parental disapproval of Marijuana
- Religious engagement
- Socioeconomic factors

Future Work & Improvements

- Incoporate other predictors such as:
  - Mental health(are there any other inducing factors)
- Use more advanced resampling techniques such as SMOTE
- Obtain dataset that is more representative of the classes.

Thank You!